

activating the complementary pair of MOSFETs T2 and T4. Thereafter, the circuit remains latched in this state unless over-current or low-battery conditions are encountered. This will become clear to you once you go through further description of the circuit that follows.

In the circuit, op-amp LM358(A) is connected as a high-side current monitor. The differential input for this op-amp is the sense voltage ($V_{\text{sense}} = I_{\text{Load}} \times R_{\text{sense}}$) developed across the 0.05-ohm resistor (R_{sense}), which is connected across points B and E in Fig. 5. Output voltage V_o developed across resistor R_b is given by the relationship:

$$V_o = V_{\text{sense}} \times R_b / R_a$$

(In the circuit, $R_a = 1$ kilo-ohm and $R_b = 8.2$ kilo-ohms)

If we assume the maximum permissible load current as 8A, then:

$$V_{\text{sense}} = 8 \times 0.05 \text{V} = 0.4 \text{V}$$

$$\text{and } V_o = 0.4 \times 8.2 = 3.28 \text{V}$$

Thus if you want the load to trip when current exceeds 8A, you should set the inverting pin of LM358(B) (connected as Schmitt trigger) at 3.28V. When the output voltage across R_b (connected to its non-inverting pin) exceeds 3.28V, a positive pulse is generated at its output. This positive pulse sets comparator N1 (connected as a bistable) to high output state. (Note that N1 is initially reset to low output state because of slightly positive bias applied at its inverting pin with the help of resistive voltage divider formed by resistors R15 and R16). The high output of N1 is inverted by gate N2. Irrespective of the output state of comparator N2, point D is pulled low to switch off the load. Thus once overload occurs, the battery is

disconnected from the load.

Unless the cause of overload is traced and fault removed, you will not be able to switch on the load. Once the overload fault is rectified, you can switch on the load by momentarily pressing switch S2 (if link JS is kept open) or by pressing RESET switch S4 (if link JS is kept closed). Push switch S3 is provided to manually switch off the circuit by setting the output of bistable N1 high.

A single comparator (N3) is used for battery voltage monitoring. With link JS closed, it monitors the battery-voltage and switches the load off when the battery voltage falls below

11.25V and switches it on again when the battery voltage reaches 12.75V in accordance with the set points as referred in Fig. 2.

Regulated 5V DC, used as reference voltage, is applied to the inverting pin of comparator N3. A proportionate sample of the battery voltage is applied to the non-inverting pin of comparator N3 via potmeter VR2. At 12V battery voltage, the potmeter is set to provide 5V at the non-inverting pin. With the component values selected, you get the required switching on/off threshold points of $5V \pm 0.325V$, which corresponds to the battery voltage of $12V \pm 0.78V$ or a cut-off voltage level of 11.22V and switch-on level of 12.78V. Those interested in the mathematical analysis may follow the relationships given in Fig. 7.

The output of comparator N2 is used for load-off indication. It lights up LED2 when the load is cut-off. The outputs of N2 and N3 form a wired-OR junction. Thus if any of these comparator outputs is low, the output at the junction point is low. Conversely, both the outputs need to be high so that the output at the junction is high.

Overload-current comparator N1 acts as a bistable latch. Hence once set to high state due to overload, the reset button (switch S4) must be pressed to reset it after removal of any fault that caused the overload. To drive the MOSFET (T4 in Fig. 5), a high rate of change of current for gate drive is needed.

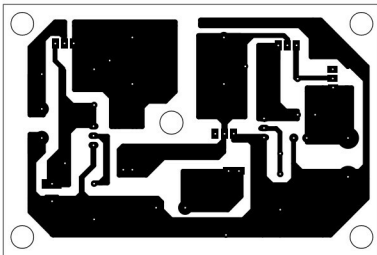


Fig. 11: An actual-size, single-side PCB for the main charger circuit shown in Fig. 5

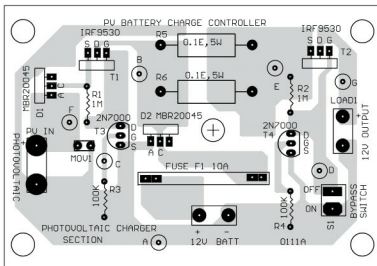


Fig. 12: Component layout for the PCB in Fig. 11